

Beyond drain amylase: expanded biological monitoring in the service of diagnosis

-Case Report-

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Introduction

Monitoring of pancreatic fistulas is classically based on repeated drain amylase measurements, with normalization interpreted as a sign of resolution. However, this approach may miss low-grade progressive complications, since a single biomarker cannot capture the full biological trajectory of an evolving fistula, particularly when clinical signs remain discreet or non-specific. This discrepancy between a reassuring biochemical value and an unrecognized clinical trajectory underscores the limitations of relying on a single, static biomarker for post-operative surveillance.

Objective

Our study aimed to demonstrate the contribution of an expanded biological monitoring panel and its kinetic analysis for the early detection of complications following pancreatic fistula.

Observation

We report the case of the patient M.A., hospitalized in the intensive care unit. A longitudinal biological assessment was performed repeatedly, including drain amylase, renal function (urea, creatinine), hepatic parameters (AST,ALT,ALP, GGT, bilirubin), inflammatory markers (CRP), albuminemia, and blood electrolytes (sodium; potassium and calcium).

Results and Discussion

The patient's biochemical profile revealed a striking dissociation. Over the first six weeks, drain amylase decreased markedly from 134,670 U/L to 60 U/L, suggesting successful fistula closure. Over the following three weeks, however, a paradoxical and progressive multiorgan biological deterioration emerged, characterized by worsening renal function (urea 0.32→1.49 g/L; creatinine 5→37 mg/L) and a predominantly cholestatic rather than hepatocellular pattern of liver dysfunction: cytolysis markers (AST/ALT) remained within normal limits throughout, whereas ALP (147→469 U/L), GGT (68→213 U/L), and total bilirubin (7→15 mg/L, predominantly conjugated) progressively rose. This hepatorenal deterioration occurred against a background of persistent systemic inflammation (CRP 55–144 mg/L) and hypoalbuminemia (27–34 g/L), accompanied by electrolyte disturbances that further reflected the underlying multiorgan dysfunction: progressive hypernatremia (up to 155 mmol/L) consistent with uncompensated free-water loss in an inflammatory state, a tendency toward hypocalcemia suggestive of fat saponification from peripancreatic necrosis, and fluctuating potassium levels tracking with worsening renal function.

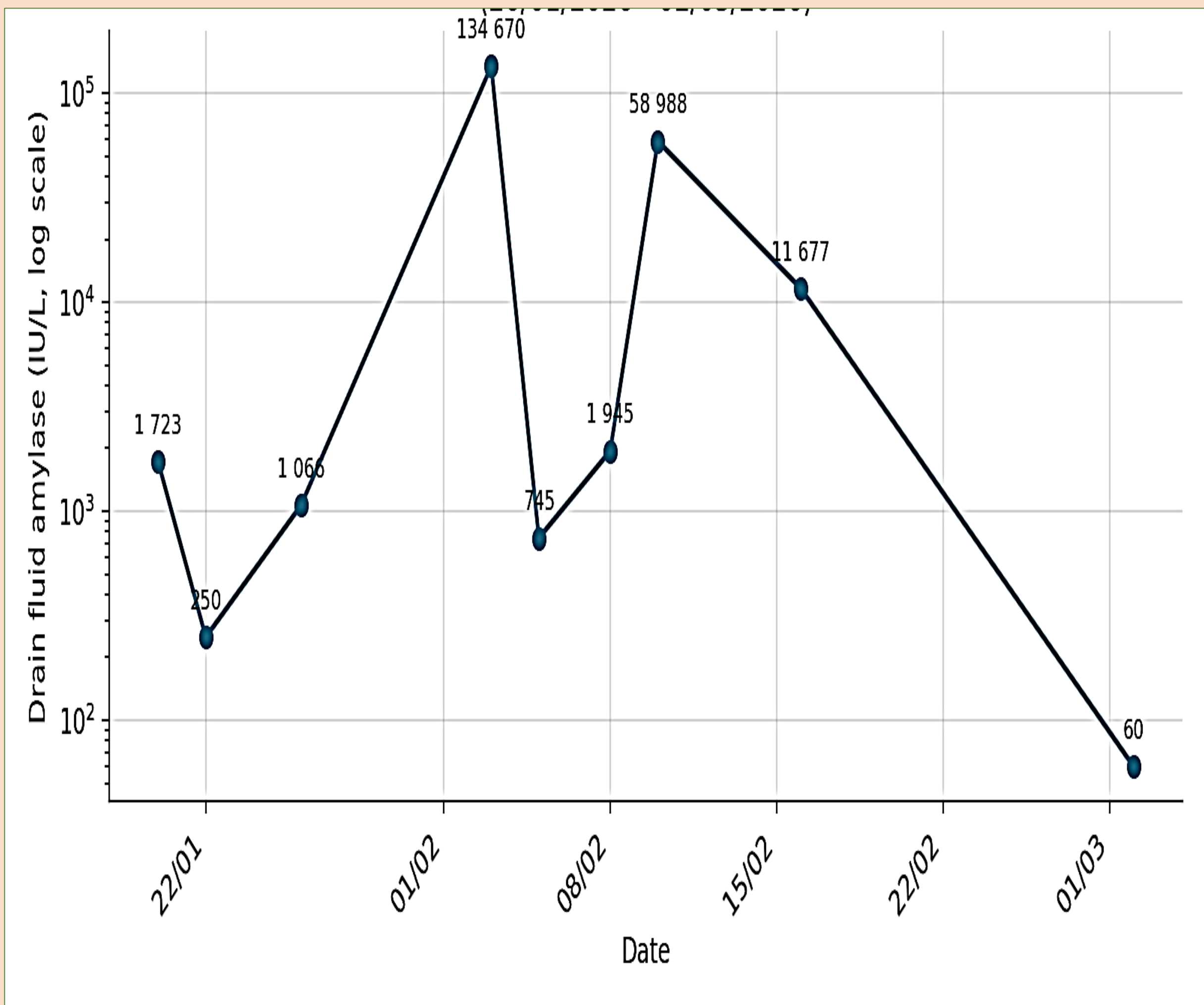


Figure 1: EVOLUTION OF DRAIN FLUID AMYLASE

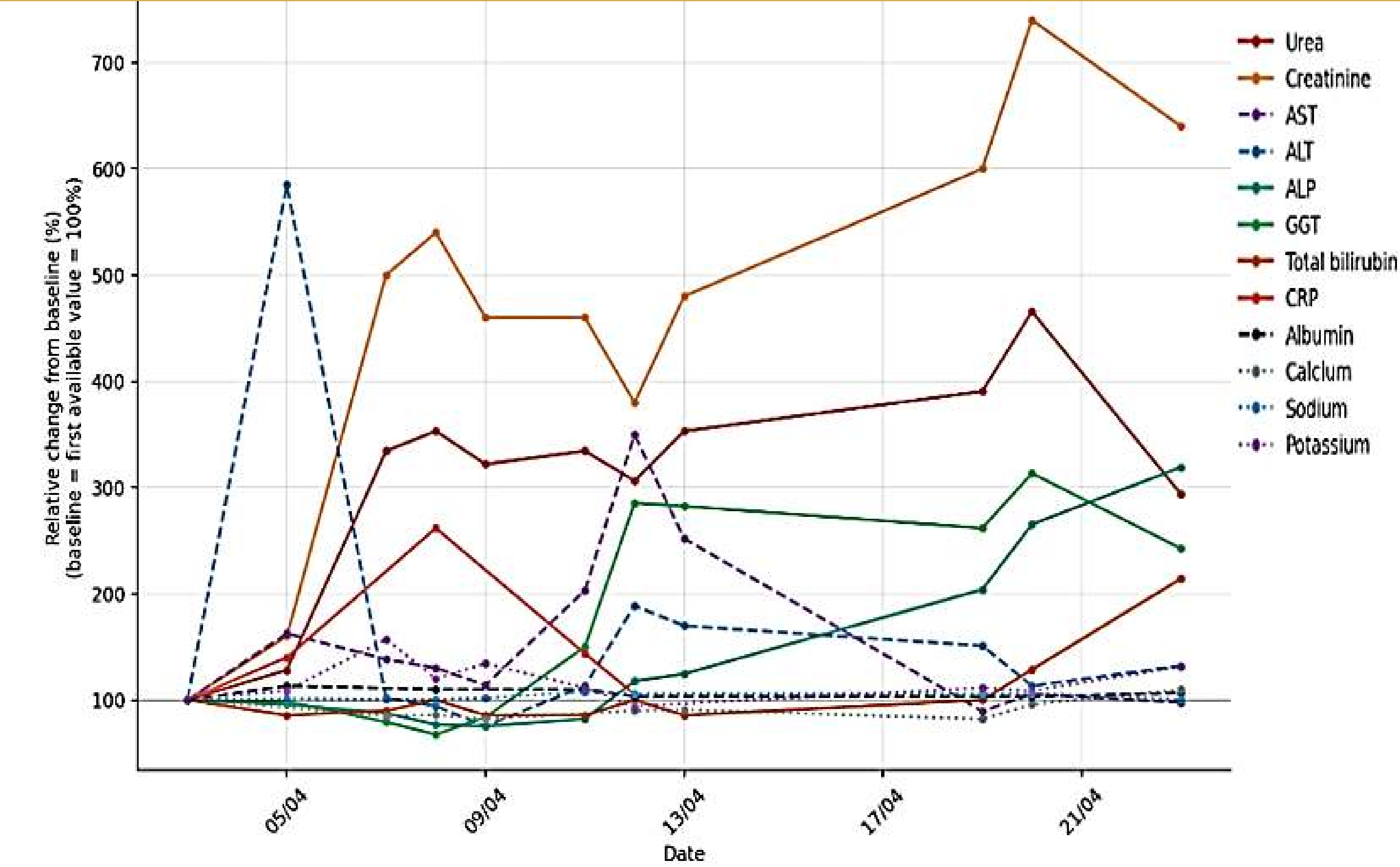


Figure 2: EVOLUTION OF BIOLOGICAL PARAMETERS

Conclusion

Normalization of drain amylase is insufficient to exclude a secondary complication, as this case shows biochemical resolution of the fistula preceding, not excluding, later multi-organ deterioration. Maintaining an expanded biological workup and its joint kinetic interpretation, rather than reliance on a single normalizing marker, appears essential for early detection and appropriate diagnostic guidance.

Bibliographic reference

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